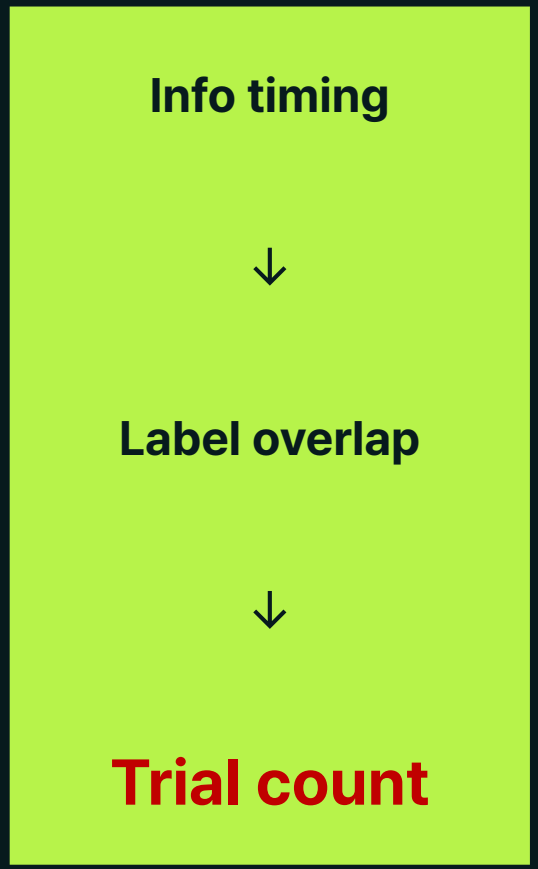
Honest Backtesting
Not Fooled by the Past

Does a strategy that earned well really work?



Financial ML · Week 10

This week runs in four steps plus one extension

From basics to core concepts, cases, then summary

01	**Basics**	**Tools to read a backtest** Returns, train/test, overfitting
02	**Leakage, overlap**	**Use only what was known then** Common errors, purging, walk-forward
02+	**Labels and CV**	**What to predict and how to test it** Triple barrier, meta-labels, purged CV, CPCV
03	**Luck, multiple tests**	**Many trials yield lucky winners** Sharpe ratio, best Sharpe by luck, DSR
04	**Cases, summary**	**What actually happened** Decay, survivorship, factor zoo

Revising against the test raises scores, not skill

01

Basics

Concepts to know before reading backtest results

Backtest metrics use three ingredients

Return, excess return, volatility: all metrics here come from these

RETURN

Return

- The % price change over a period.

EXCESS RETURN

Excess return

- Return earned above the risk-free rate.

VOLATILITY

Volatility

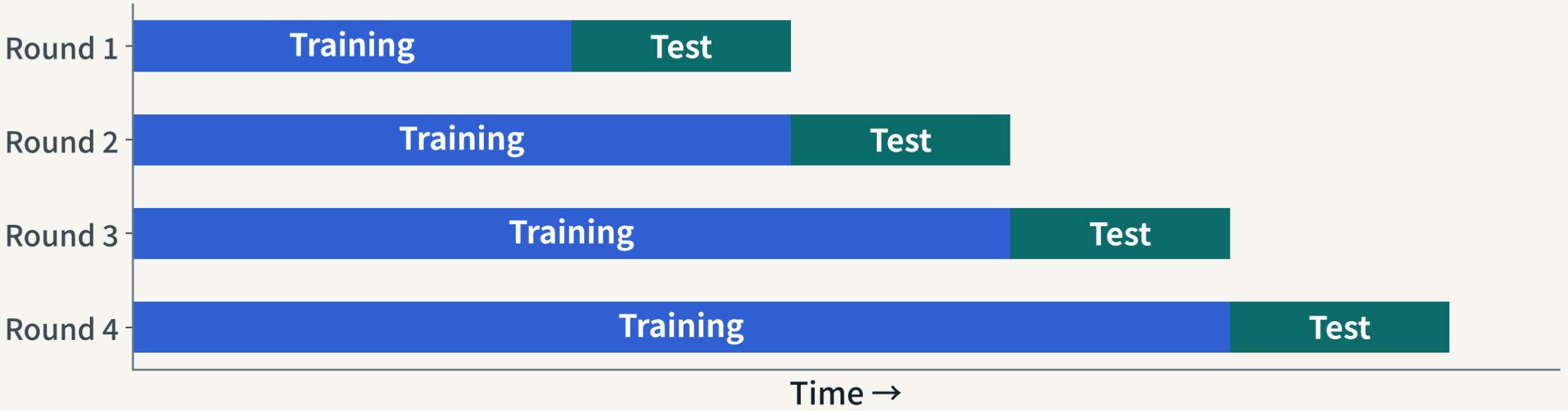
- How much returns swing around the mean (standard deviation).

Sharpe ratio is excess return divided by volatility

Mixing daily, monthly, annual units breaks numbers

Validation learns from the past, tests on the future

How training, test and out-of-sample periods relate



※ Each round trains on the past only and tests on the next period (expanding window)

The test period must be a future the model never saw

In-sample results cannot separate skill from luck

Overfitting is memorizing even past noise

Overfitting: fits training data, fails on new data

SIGNAL

Signal

- A pattern repeated across periods.
- It may persist in the future.

What we want to learn

NOISE

Noise

- A fluke of that period only.
- It vanishes in the future.

Not to be memorized

Complex models and many trials make noise look like signal

02

Leakage · Overlap · Luck

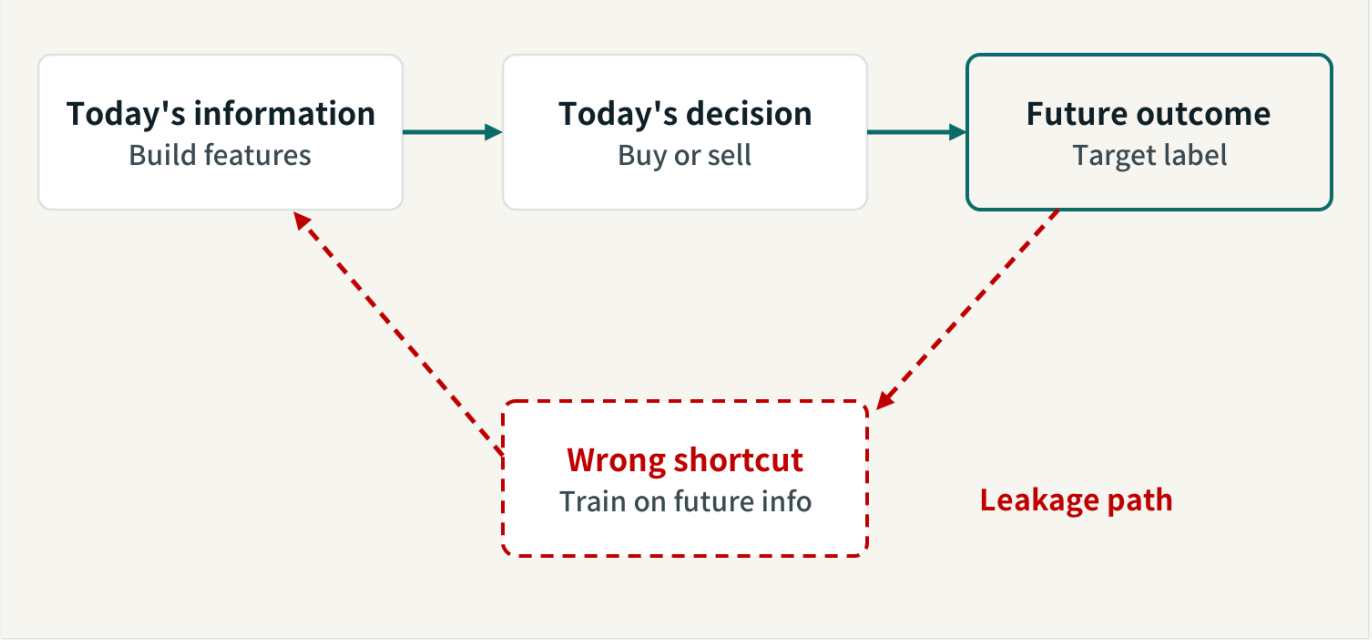
Source concepts: info timing, label spans, trials

A backtest may use only what was known at decision time

A backtest is a mock test replaying past data in time order

Look-ahead leakage

- Using earnings released after the close in that morning's trade peeks ahead.
- Standardizing with full-period mean and SD, then splitting, also leaks.



Fit scaling on training data only

Check information timing before adding model complexity

Four common mistakes all inflate performance

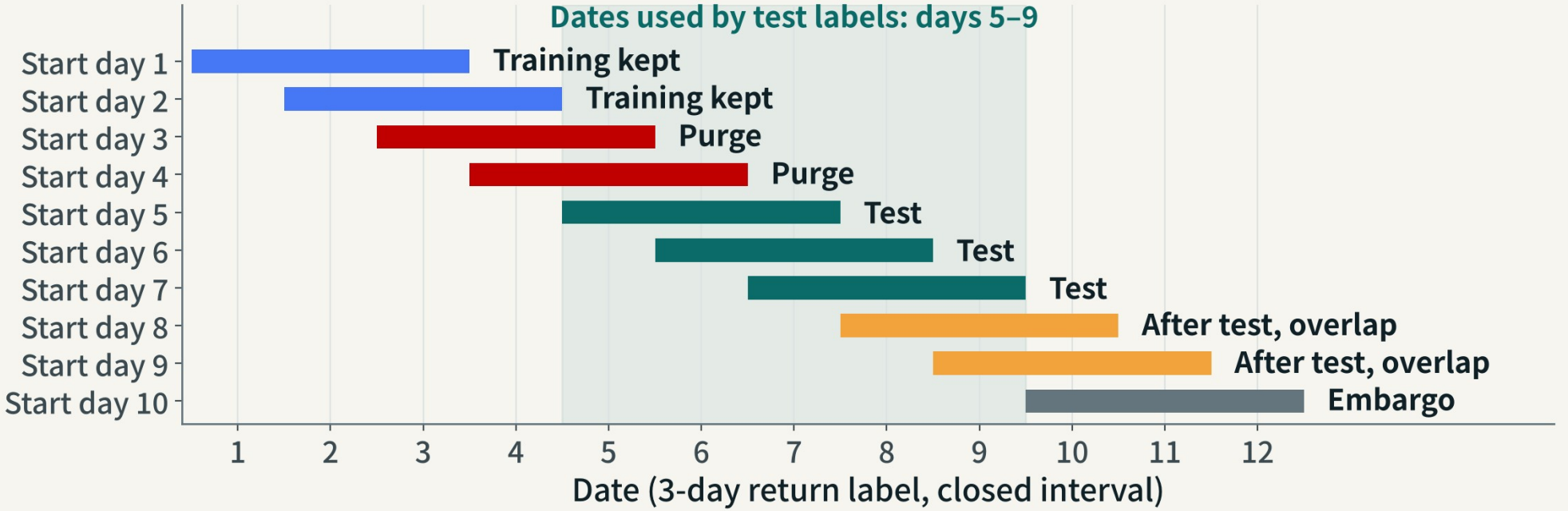
Each row: why the mistake inflates the score, and the fix

Common mistake	Why it inflates results	How to fix it
Shuffling time-series data	Future regimes help fit the past	Train on past, test on future
Surviving stocks only	Delisted failures drop out	Use stocks investable then
Preprocessing all data	Test-period traits leak into training	Fit within each training window
Revising after tests	The test set becomes training data	Split dev validation and final test

Revising against the test raises scores, not skill

Split by date, label windows can still overlap

3-day return labels, test starts days 5–7, closed intervals with endpoints



Test labels span days 5–9, overlapping training samples from days 3–4

Compare each sample's full window, not its start date

Purging and time order are separate constraints

'Purged CV' does not mean 'always train on the past only'

PURGING

Purging

- Drop training samples overlapping test info or label windows.

EMBARGO

Embargo

- Also drop some samples after the test window.
- Set its length to the data and info window.

WALK-FORWARD

Walk-forward

- Train on the past, test on later periods.
- Replicates the time order of live trading.

Some Purged CV also trains on data after the test window

※ Source note: the p.13 date example is simplified around start dates; do not generalize

Sharpe ratio is return divided by fluctuation

Compute the Sharpe ratio in one consistent period unit

Definition

$$SR = \frac{\text{mean excess return}}{\text{std. dev. of excess return}}$$

E.g. monthly mean excess return 1%, monthly vol. 2%

$$\text{Monthly Sharpe } \frac{1\%}{2\%} = 0.5 \implies \text{annualized } 0.5\sqrt{12} \approx 1.73$$

Annualizing assumes independent, stable variance. Do not use it as is under strong autocorrelation

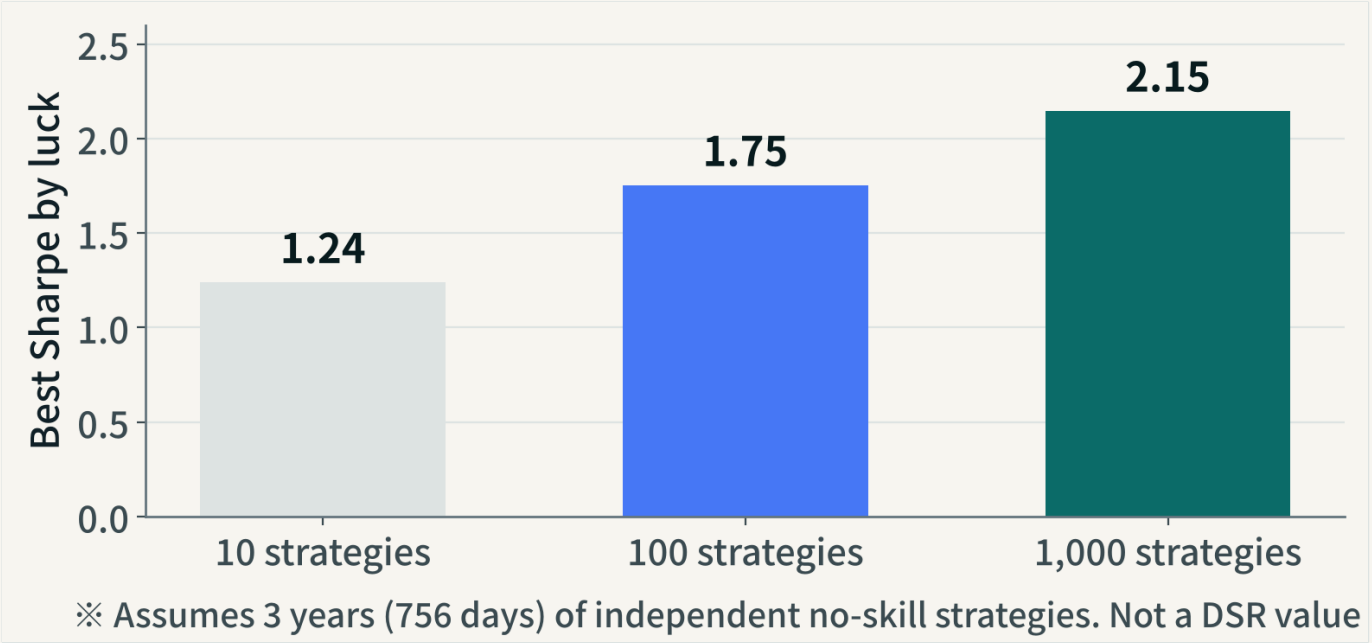
※ Source p.8 divides variance by 5 for teaching. Sample SD (denominator 4) gives another value

Many trials yield good strategies by luck

More trials raise the top Sharpe even with no skill

Multiple testing

- Among 1,000 coin-flip strategies, some win streaks by luck.
- With 1,000 tests at 5%, about 50 look significant even if all are null.



Reporting only the winner overstates skill

The expected 50 needs no independence, but the winner's distribution does

The luck-only top Sharpe is a rough benchmark

1,000 trials on 3 years (756 days) reach about 2.15 with no skill

Source's rough formula

Approx. best Sharpe by luck $\approx \sqrt{252}\,\frac{\sqrt{2\ln N}}{\sqrt{T}}$, $N =$ number of trials, $T =$ number of daily obs

Limits

- An intuitive value for independent no-skill trials; not DSR itself.

DSR (Deflated Sharpe Ratio)

- Judges observed Sharpe given trials, sample length, skew, kurtosis and dependence.

Do not read DSR 0.95 as '95% posterior probability of skill'

02+

Labels and cross-validation

Adds what to predict (labels) and how to test it (cross-validation)

Seven new terms, defined first

Terms used in this section — read them with these meanings

Label	The answer a model must predict, e.g. up (+1) or down (−1)
Fixed-horizon label	Labels by the return after a fixed period (e.g. 10 days) only
Triple-barrier label	Labels by whichever is hit first: upper (take profit), lower (stop loss) or vertical (time limit) barrier
Meta-labeling	A secondary model predicts whether to act on the side chosen by a primary model
K-fold CV	Splits data into K groups and uses each group once as the test set
Purged K-fold	K-fold that purges training samples overlapping test labels and adds an embargo after the test
CPCV	Combinatorial Purged CV: tests on combinations of groups and builds several backtest paths

A fixed-horizon label ignores the path

Looking only at the 10-day return hides stop-outs and profit-taking on the way



Up 2% after 10 days, so 'up' — yet a real trade stopped out at −3% on day 4

A fixed ±2% threshold stays the same whether volatility rises or falls

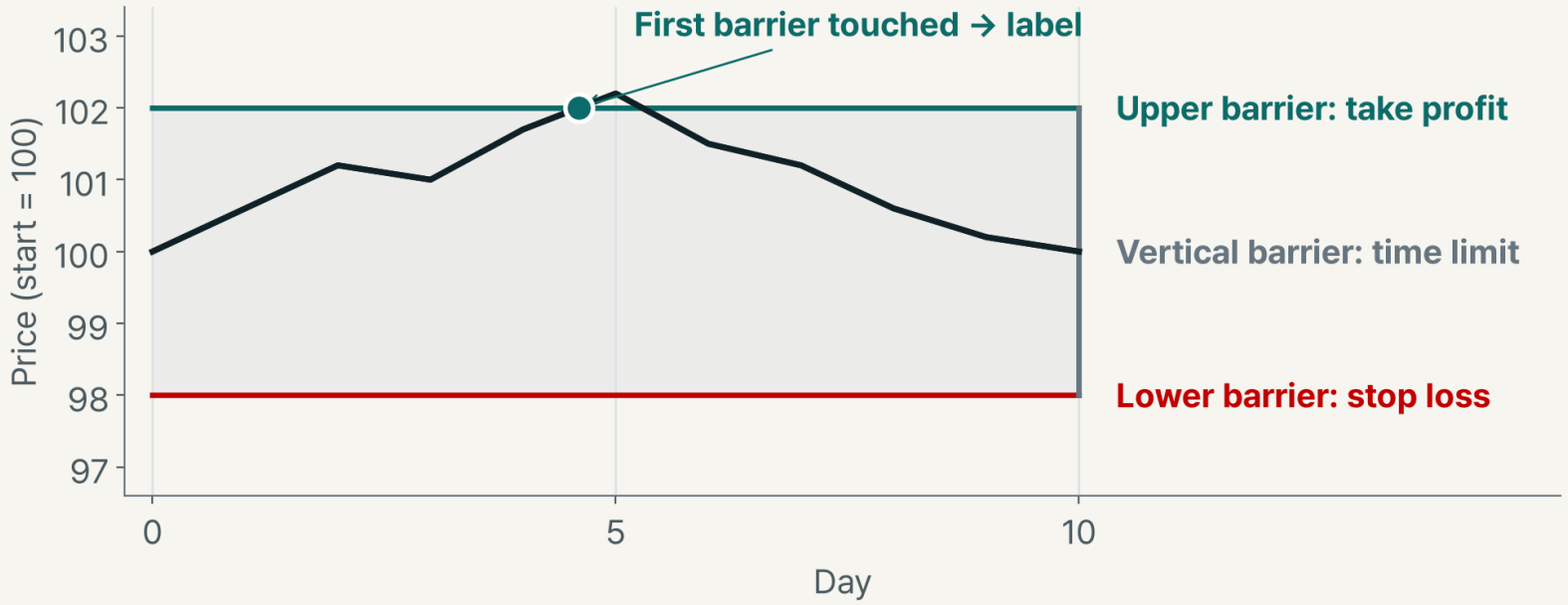
The triple barrier labels by the first barrier touched

López de Prado (2018), ch. 3 — mirrors a real trade's take-profit, stop-loss and time limit

Upper barrier: take profit

Lower barrier: stop loss

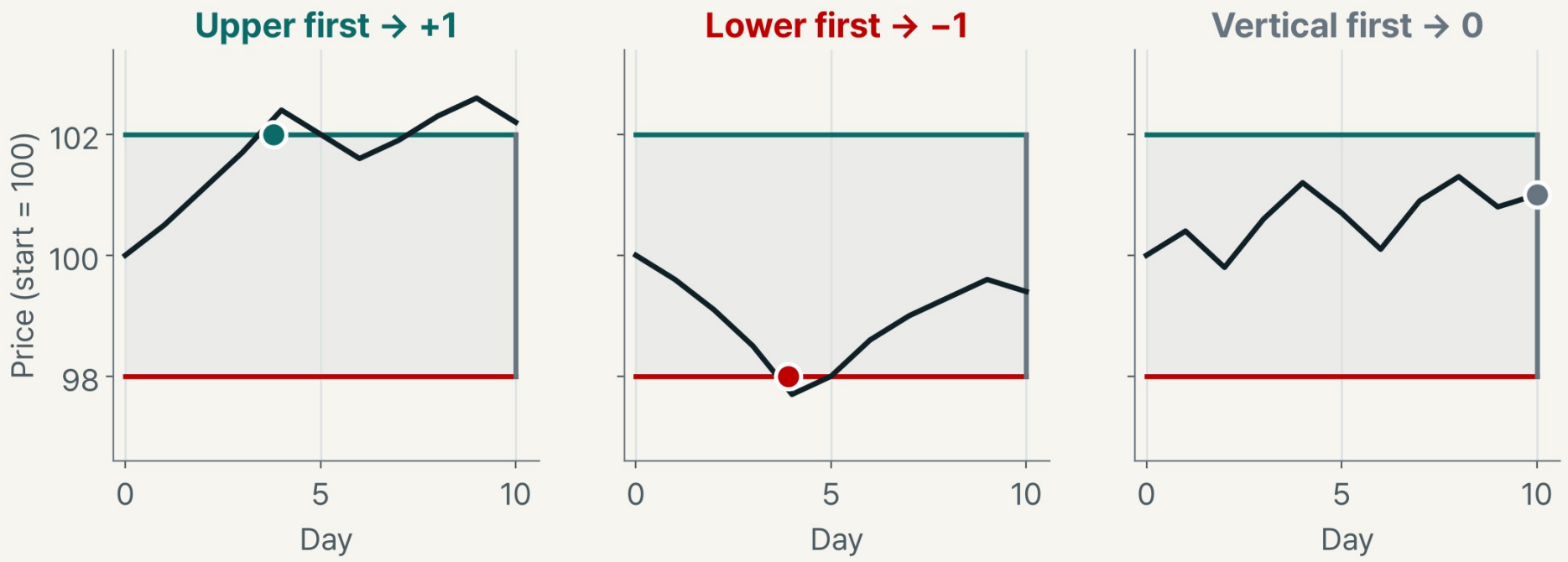
Vertical barrier: time limit



Barrier width scales with volatility — narrow when calm, wide when rough

Outcomes split three ways

Same three barriers, different price paths — the first barrier touched is the label

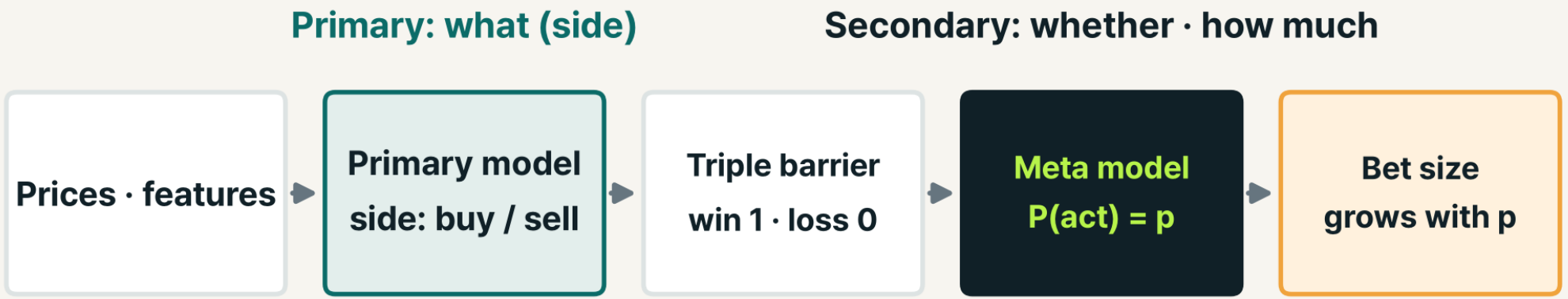


Upper +1 · lower −1 · vertical 0

At the vertical barrier, 0 or the sign of the return is a design choice

Meta-labeling separates 'side' from 'whether to act'

The primary model sets the side; the secondary model gives the probability of acting on it



Triple-barrier outcomes label each primary signal win (1) or loss (0)

The meta model's probability p sets whether to trade and the bet size

Meta-labeling raises precision at some cost in recall

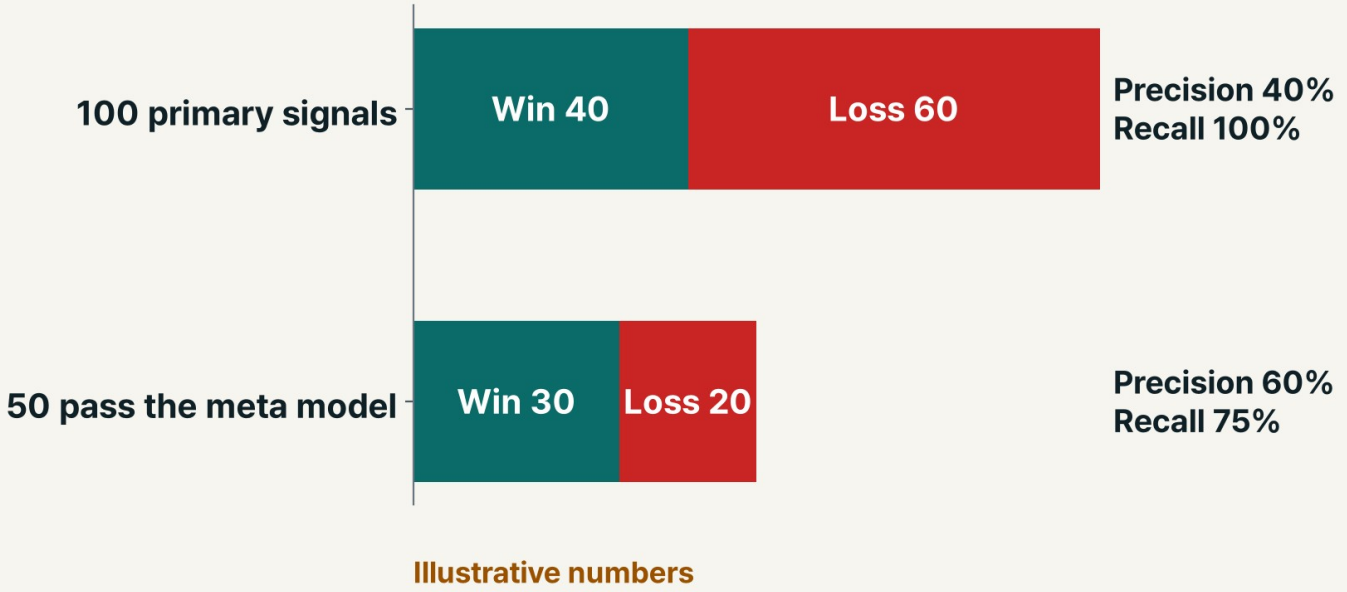
Filters primary signals to keep trades more likely to work (illustrative numbers)

Precision

Share of trades taken that worked

Recall

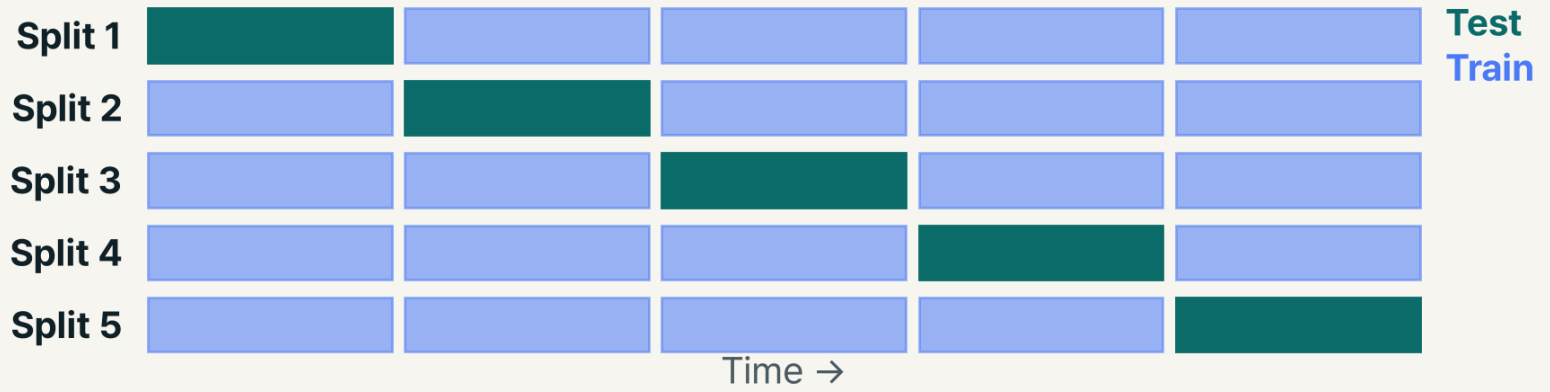
Share of possible wins actually traded



Primary model: catch chances (recall). Meta model: filter bad ones (precision)

Standard K-fold leaks test information in financial data

K-fold CV — each group takes a turn as test, the rest train





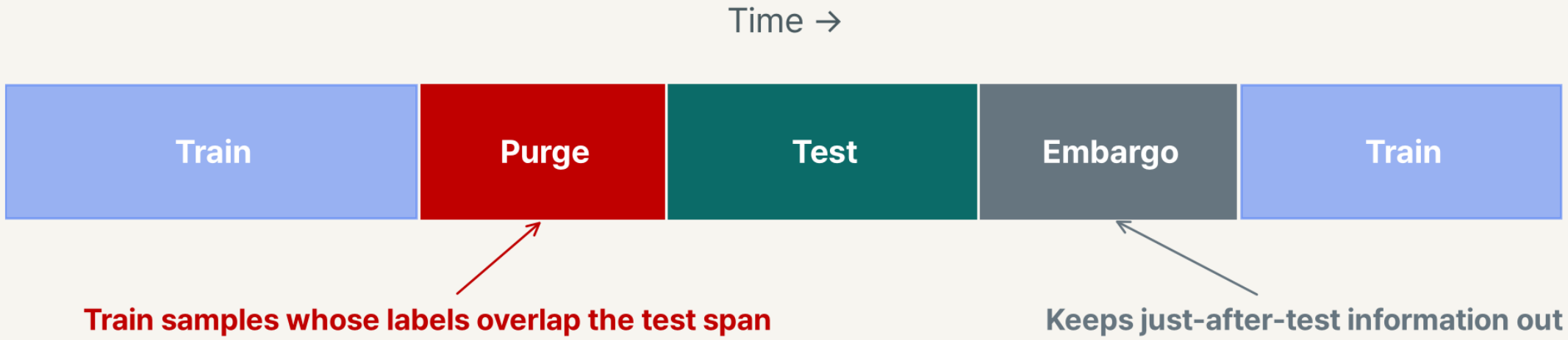
Shuffled K-fold: days right next to a test day land in training

Neighbouring days are alike and label spans overlap, so test answers leak

Shuffling makes it worse: days next to test days land in training

Purged K-fold removes overlaps and adds an embargo

López de Prado (2018), ch. 7 — purging and embargo



Drop training samples overlapping the test span, plus some right after it

Size the embargo to how long information spreads — the source example uses about 1% of the sample

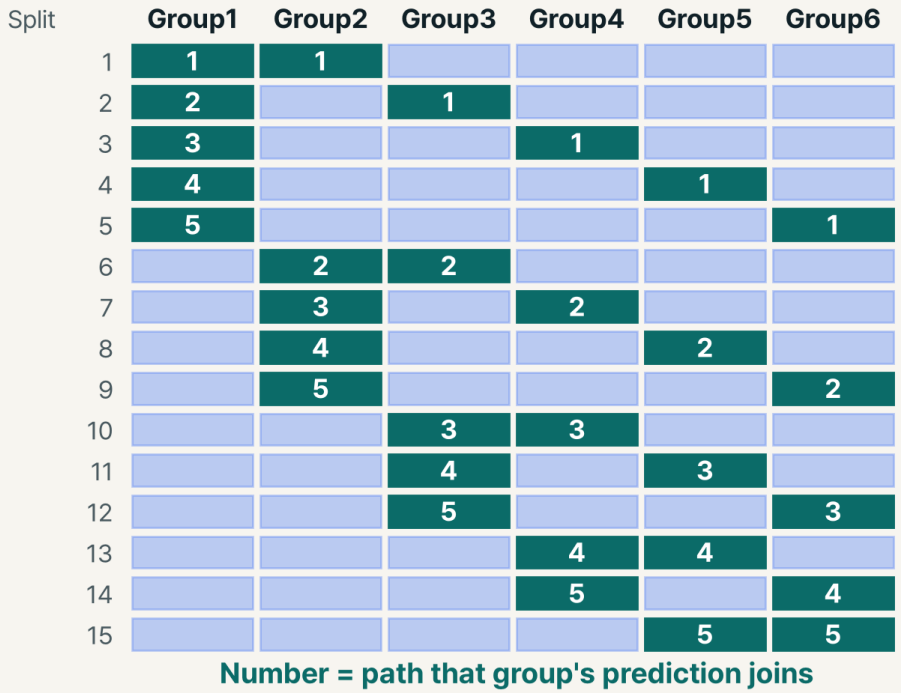
CPCV tests group combinations and builds many paths

N = 6 groups, test k = 2 at a time → C(6,2) = 15 splits

Splits: C(N, k) = C(6, 2) = 15

Each group is tested C(N−1, k−1) = 5 times

Paths: φ = k/N · C(N, k) = 5



Purging and embargo still apply inside every split

CPCV gives a distribution, not a single backtest line

Joining the same path numbers gives 5 backtest paths built only from out-of-sample predictions

	Group1	Group2	Group3	Group4	Group5	Group6
Path 1	Split 1	Split 1	Split 2	Split 3	Split 4	Split 5
Path 2	Split 2	Split 6	Split 6	Split 7	Split 8	Split 9
Path 3	Split 3	Split 7	Split 10	Split 10	Split 11	Split 12
Path 4	Split 4	Split 8	Split 11	Split 13	Split 13	Split 14
Path 5	Split 5	Split 9	Split 12	Split 14	Split 15	Split 15

One backtest line: each group's test prediction used once

→ Sharpe per path → read as a distribution

5 paths → 5 Sharpe ratios → read the range of this strategy's Sharpe

Walk-forward has one path, so luck and skill are hard to tell apart

Four validation methods in one table

What each protects, and what it returns

Method	Time order	Label overlap	Result
Shuffled K-fold	Mixed	Leaks	One score (inflated)
Purged K-fold + embargo	Partly	Purged	One score
Walk-forward	Kept	Purge at boundary	One path
CPCV	Partly	Purged	Many paths → distribution

Use walk-forward to mimic live order, and CPCV alongside it to gauge overfitting

Four questions check the extension

One question, one answer

What is wrong with fixed-horizon labels?	They ignore the path (stops, profit-taking) and the threshold ignores volatility.
How is a triple-barrier label set?	By the first of the upper, lower or vertical barrier; widths scale with volatility.
What does the meta model predict?	Whether to act on the primary signal (1) or not (0); its probability sizes the bet.
Why CPCV over walk-forward?	Several paths turn Sharpe into a distribution, so overfitting can be gauged.

※ Source: López de Prado, M. (2018), Advances in Financial Machine Learning, ch. 3, 7, 12. Price paths and counts in figures are illustrative

03

Lessons from cases

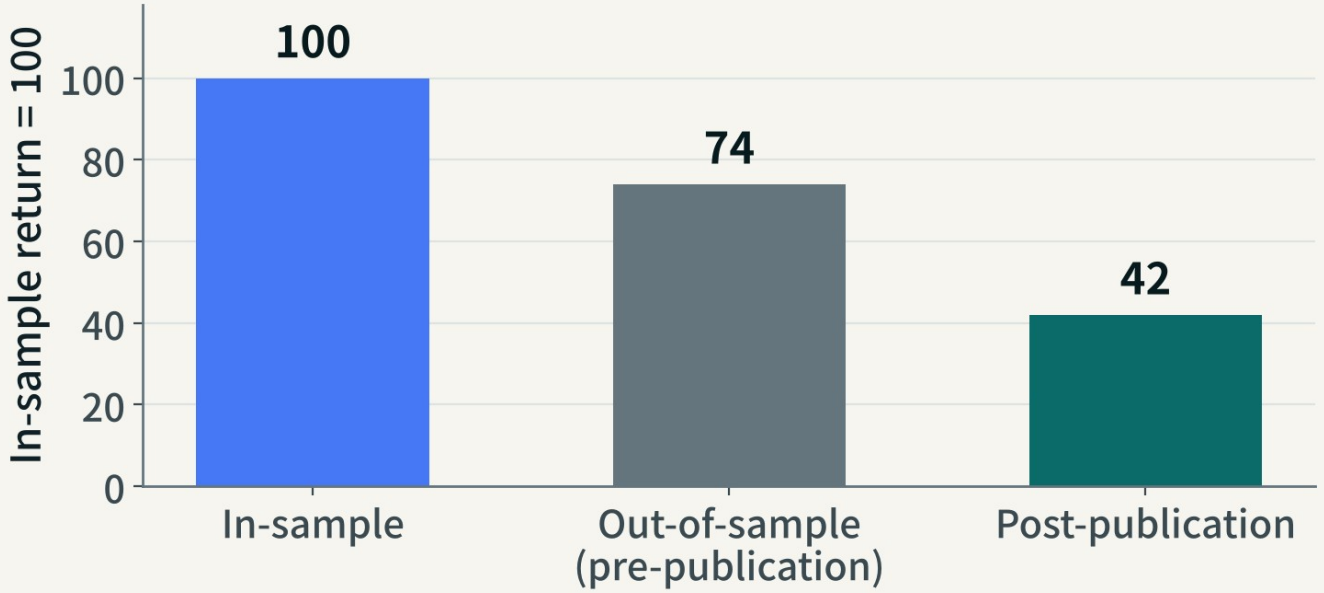
What academia and industry actually observed

Published signals lost much of their return after publication

McLean and Pontiff (2016, Journal of Finance): 97 return predictors tracked

What they found

- Returns fell about 26% after the sample period.
- Returns fell about 58% after publication.



※ McLean & Pontiff (2016), 97 predictors. 26% and 58% declines as index

Partly overfitting, partly investors copying it

Read backtest results as an upper bound on the future

Seeing only survivors makes the past look better

Survivorship bias: failed samples vanish from the data

Bomber story

- In WWII, bullet holes on returning bombers pointed to the wrong spots to armor.
- Planes that never returned left no data.

Fund data

- Dropping closed funds inflates average returns.

Index members

- Using today's members drops delisted stocks.

Backtest on the full universe investable at the time

Hundreds of reported factors: a classic multiple-testing case

Factor zoo: with enough trials, some factors look significant

Harvey·Liu·Zhu (2016)

- Over 300 return factors had been reported.
- They proposed a stricter t ≥ 3.0 hurdle for new factors.

Butter and S&P 500

- Leinweber ‘explained’ the S&P 500 with Bangladesh butter output.
- Data mining finds absurd links.

Lesson

- Record and disclose trial counts.
- Test economically grounded ideas first.

Check not just the winner's score but how many tries it took

04

Summary

Close the week with key Q&A

This week: three checks on information, overlap and luck

One question, one answer for this week

Check question	Answer
Is a time-ordered split enough?	**No. Also check label overlap, preprocessing leakage, test reuse.**
Does high Sharpe prove skill?	**Only after checking trials, data length, costs and leakage.**
Are purging and walk-forward the same?	**No. One removes overlap; the other enforces time order.**
Is DSR 0.95 a 95% skill chance?	**No. It is statistical evidence under stated assumptions.**

Honest validation before fancy models

WHEN DID YOU KNOW IT?

HOW MANY TIMES DID YOU TRY?

THEN TRUST THE SHARPE.

Report decision-time info
and trial counts together.